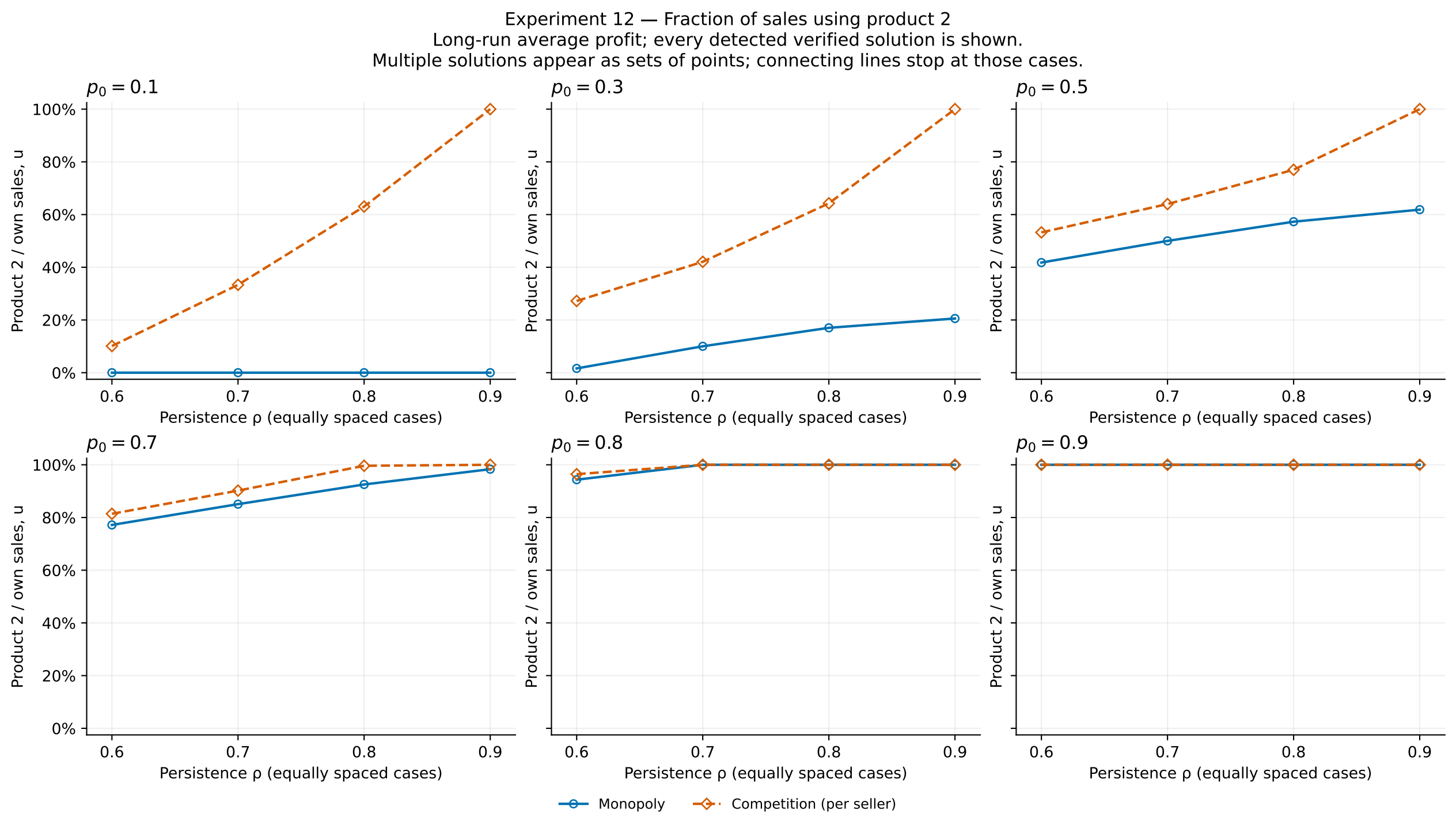
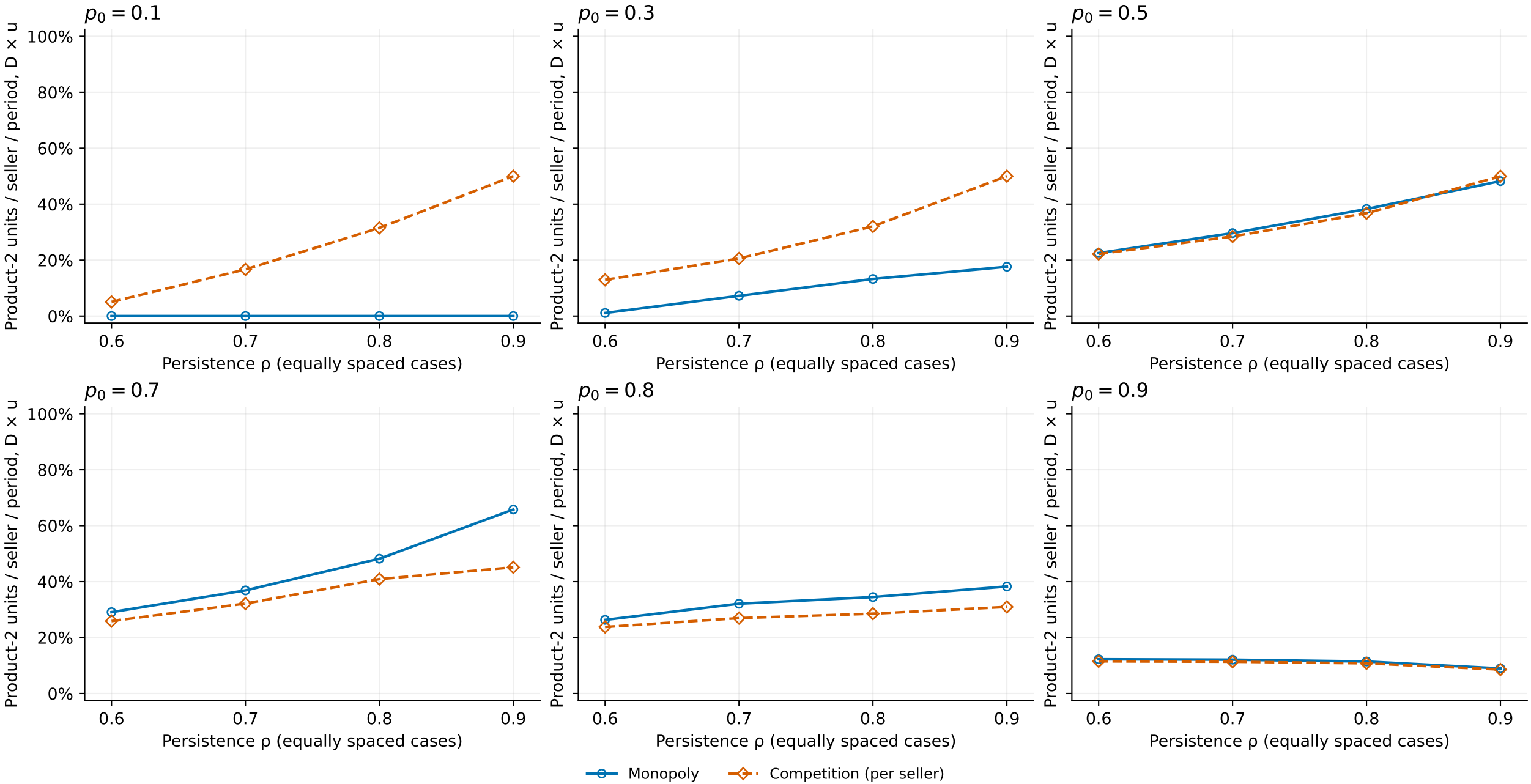
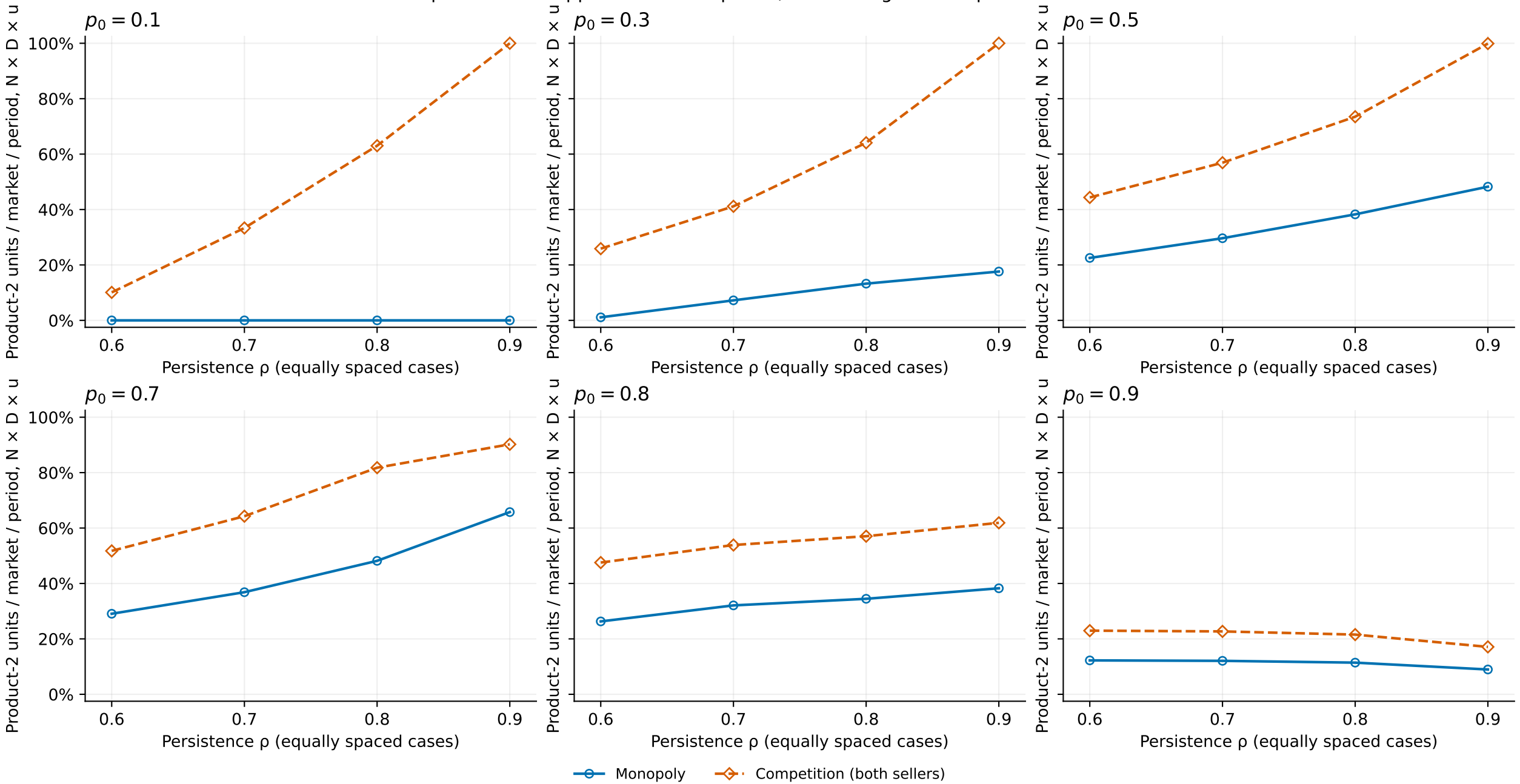
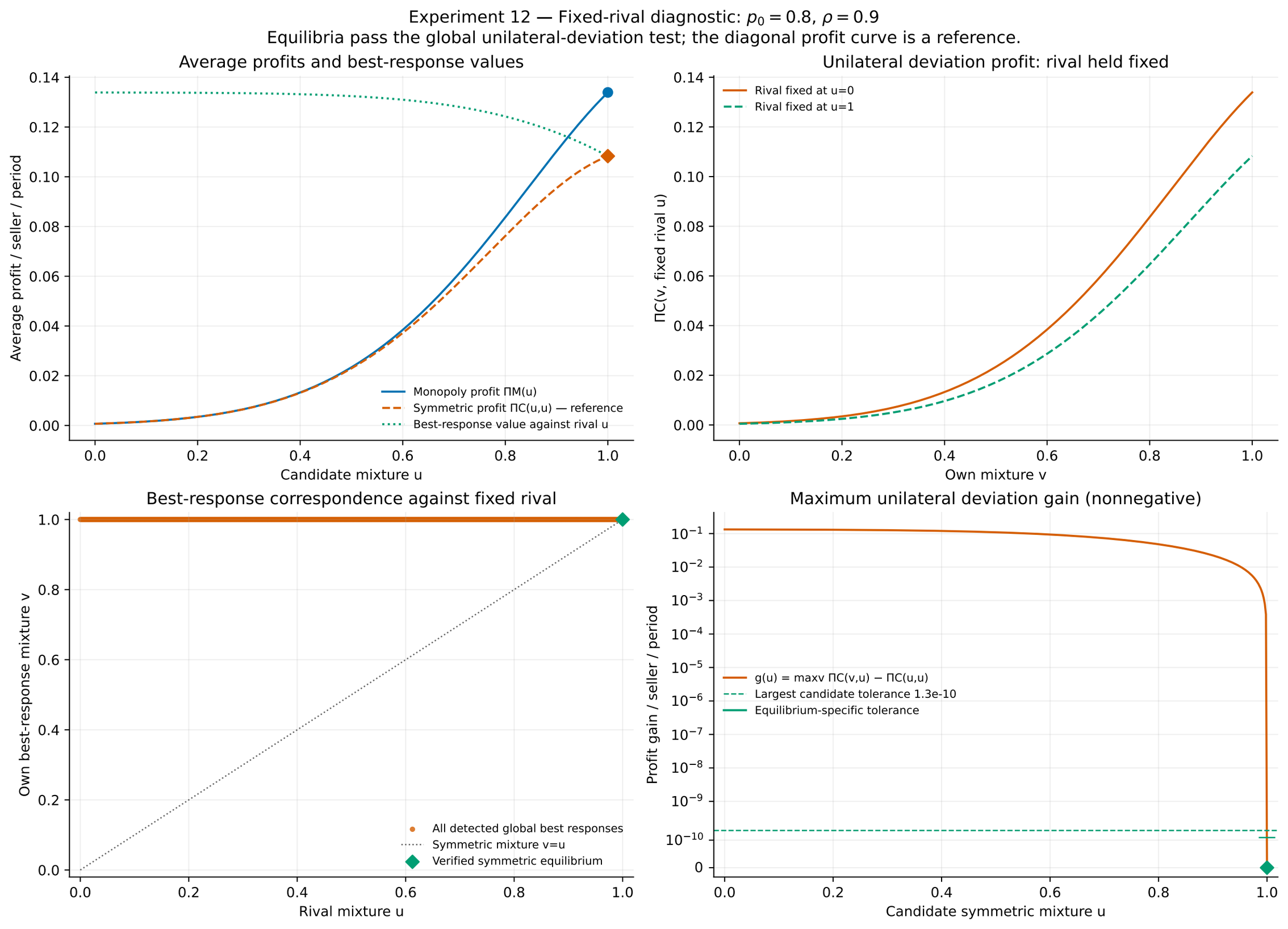

Experiment 12 — Fraction of sales using product 2
Long-run average profit; every detected verified solution is shown.
Multiple solutions appear as sets of points; connecting lines stop at those cases.
$p_0 = 0.1$
$p_0 = 0.3$
$p_0 = 0.5$
$p_0 = 0.7$
$p_0 = 0.8$
$p_0 = 0.9$
Product 2 / own sales, u
0%
20%
40%
60%
80%
100%
0.6
0.7
0.8
0.9
Persistence ρ (equally spaced cases)
Monopoly
Competition (per seller)


Experiment 12 — Product-2 use per seller and per period
Long-run average profit; every detected verified solution is shown.
Multiple solutions appear as sets of points; connecting lines stop at those cases.
$p_0 = 0.1$
$p_0 = 0.3$
$p_0 = 0.5$
$p_0 = 0.7$
$p_0 = 0.8$
$p_0 = 0.9$
Product-2 units / seller / period, D × u
100%
80%
60%
40%
20%
0%
0.6
0.7
0.8
0.9
Persistence ρ (equally spaced cases)
Monopoly
Competition (per seller)


Experiment 12 — Total product-2 use in the market per period
Long-run average profit; every detected verified solution is shown.
Multiple solutions appear as sets of points; connecting lines stop at those cases.
$p_0 = 0.1$
$p_0 = 0.3$
$p_0 = 0.5$
$p_0 = 0.7$
$p_0 = 0.8$
$p_0 = 0.9$
Product-2 units / market / period, N × D × u
Persistence ρ (equally spaced cases)
0%
20%
40%
60%
80%
100%
0.6
0.7
0.8
0.9
Monopoly
Competition (both sellers)


Experiment 12 — Fixed-rival diagnostic: $p_0 = 0.8$, $\rho = 0.9$
Equilibria pass the global unilateral-deviation test; the diagonal profit curve is a reference.
Average profits and best-response values
Average profit / seller / period
Candidate mixture u
Monopoly profit ΠM(u)
Symmetric profit ΠC(u,u) — reference
Best-response value against rival u
Unilateral deviation profit: rival held fixed
ΠC(v, fixed rival u)
Own mixture v
Rival fixed at u=0
Rival fixed at u=1
Best-response correspondence against fixed rival
Own best-response mixture v
Rival mixture u
All detected global best responses
Symmetric mixture v=u
Verified symmetric equilibrium
Maximum unilateral deviation gain (nonnegative)
Profit gain / seller / period
Candidate symmetric mixture u
g(u) = maxv ΠC(v,u) − ΠC(u,u)
Largest candidate tolerance 1.3e-10
Equilibrium-specific tolerance